# OCD Automation Architecture Assessment

**Custom supporting system and n8n comparison**

Prepared 24 October 2026 · OCD internal operations and future branches

**Recommendation:** Make the existing custom application responsible for the core automation lifecycle. Keep ShiftCare authoritative for care operations and Xero authoritative for accounting. n8n is optional for a separately justified peripheral connector or notification step; it is not a required production dependency.

The important work is matching the correct person and occurrence, applying reviewed rules, maintaining ownership, checking external outcomes and recovering safely. The existing code already provides part of this foundation. Adding a second primary engine would duplicate responsibilities without removing those requirements.

**Coverage:** 96 pain points, the 14 established workflow families, current custom code, the inactive n8n onboarding draft, prior research and existing workflow diagrams. The complete per-item assessment follows the main report and is also available in `opportunity-matrix.csv`.

| Treatment of the 96 items | Items |
|---|---|
| A — Automate bounded preparation or coordination | 19 |
| R — Automate preparation with a reviewed business action | 38 |
| C — Clarify the rule or meaning before configuration | 8 |
| N — Use or configure the native platform | 4 |
| H — Mainly human, process or IT work | 27 |

These are assessment categories, not 96 separate software projects or measured savings. Related items share services and workflows. Source descriptions, proposed treatment, human boundaries and completion evidence remain distinct in the matrix.

**Reading order:** decision and opportunities (§§01–05); objective comparison (§§06–07); custom architecture and ownership (§§08–09); implementation gaps (§10); costs and rollout (§§11–12); optional n8n (§13); evidence (§§14–15); full opportunity matrix (appendix).

# 01 Recommendation and scope

**A custom supporting system can handle the required software workflows without n8n.** It needs durable jobs, schedules, integration clients and operational controls; these are ordinary application components. n8n packages many of them, but is not a prerequisite for using the vendors' APIs.

For OCD, custom ownership is practical because the work already combines an office review interface, domain-specific policy, sensitive records, approvals, source changes and potentially uncertain external writes. The current backend implements meaningful controls for that lifecycle. The n8n draft in the repository is inactive; its artifacts do not demonstrate a completed production integration chain.

This revises the earlier default hybrid recommendation. A second workflow engine is worthwhile only when a particular connector or independently maintained peripheral process saves more effort than its additional operation and integration cost.

## The boundaries

- **ShiftCare:** authoritative people, roster, acceptance, attendance, care entry and native documents. Use existing native capabilities before reproducing them.
- **Xero:** authoritative accounting contacts, invoices, credits and payments. Payroll remains with the confirmed payroll product; Xero accounting use does not prove Xero Payroll use.
- **Custom system:** sources and their revisions, cases, reviewed proposals, approvals, jobs, schedules, external-ID links, communication handoffs, exceptions and verified outcomes.
- **People:** capacity decisions, identity ambiguities, agreed financial treatment, employment decisions, care judgement, real service delivery and unresolved exceptions.

The architecture assumes a named developer/maintainer and funded operational ownership. Without that, a custom system can become a new owner dependency. n8n also needs a maintainer, tested changes and operational support; it does not make business rules maintain themselves.

No new CRM, applicant-tracking system, payroll calculation engine, signing platform or marketing programme is proposed. The optional arrival work remains W14 and should be accepted separately after native feature fit and device/privacy testing.

# 02 Enquiries onboarding and documents

The repeated input work is a strong first target. The same source and matching services can support participant forms, worker forms, returned agreements and enquiries.

| Opportunity and pain points | Effective custom approach | Completion and human boundary |
|---|---|---|
| Enquiry ownership — PP001–008, PP013 | Authorized source ingestion; stable source IDs; area screening; required-field checks; owner and deadline. Reuse the Graph collector and shared intake. | Office resolves ambiguous people, date meaning and capacity. One source must not create duplicate cases. |
| Participant/worker onboarding — PP001–006, PP057, PP080 | Extract draft fields with provenance; show original evidence; find duplicate candidates; approve exact native fields; link checked external IDs. | Office/HR approves identity. Payroll/accounting readiness is a separate handoff; profile creation is not service readiness. |
| Agreements and filing — PP035–038, PP043 | Index originals, person/type/version/visibility checks, expected attachments and native filing references. Use the existing signing process. | Staff verify signature/version and uncertain matches. A missing native copy is not proof the agreement does not exist. |
| Dates and renewal — PP039–042 | Extract date candidates; distinguish plan, agreement and real expiry; prepare scoped completeness/renewal tasks. | Confirm ambiguous meanings and actual expiry from evidence. Correct metadata before adding more alerts. |
| Communication — PP009–012 | Reviewed templates, required context, recipient verification, approved outbox and separate delivery/response outcomes. | Current backend records native handoffs; automatic delivery is additional work. A task or API response is not an acknowledgement. |

**Why custom fits:** The same office screen must connect the source, native identity, outstanding fields and decision. Reusing these checks avoids rebuilding them in both a workflow canvas and the application.

**Where n8n helps:** It could supply a connector for an approved source that the custom code does not support, or dispatch a pre-approved notification. The existing Microsoft 365 collector means email ingestion is not, by itself, a reason to add n8n.

The actual intake date, required forms, supported native fields, document categories and acceptable extraction accuracy still need the current office team's agreement. Low-confidence extraction stays editable and reviewable.

# 03 Rostering confirmations and service follow through

Scheduling combines deterministic checks with actual worker and participant choices. Use the native roster as the source of current assignments; a custom review case coordinates the dependent steps.

| Opportunity and pain points | Effective custom approach | Human or native boundary |
|---|---|---|
| Availability and booking entry — PP015–020, PP024 | Validated request drafts; explicit timestamps; overlap/leave evidence; scope preview and native read-back. | Worker acceptance and participant agreement remain necessary. Free calendar time does not prove availability. |
| Travel and changed bookings — PP021–023 | Confirm addresses; prepare route estimates; flag invalidated travel; keep travel and service time separate. | Estimates do not become actual mileage or pay entitlement. Existing native future-series updates remain useful. |
| Shift instructions — PP025–026 | Approved versions and missing-instruction checks; preview the intended booking set before an approved update. | Confirm the supported native instruction operation. Keep a checked native handoff where unavailable. |
| Reminder replies — PP027–030 | Match to one occurrence, prepare cause/notice/travel/charge checks, route an approved action and verify it. | No response is not a cancellation. Group, financially locked and uncertain cases require native review. |
| Absences and cover — PP031–034, PP085 | List affected services; collect native candidate evidence; track offers, acceptance, participant choice and deadlines. | Staffing shortages require staffing. The system cannot invent a suitable worker or infer absence intent. |
| Care, feedback and cases — PP067–073 | Surface factual note/task/duration exceptions, feedback follow-ups and restricted evidence with an owner. | Supervisors investigate quality; managers handle personnel and safety. Existing urgent escalation remains in force. |
| Arrival enquiries — PP014 | Assess native arrival visibility; use the existing W14 concept only for a demonstrated remaining gap. | Consent, real devices, current assignment, freshness and withdrawal must be tested; no constant GPS feed is presumed. |

**Critical gap:** The current W07 handler and cancellation adapter do not implement provider-travel/mileage cleanup. Adding a reviewed no-travel precondition and verifying those lines is necessary before claiming that entire cancellation workflow is automated. Unsupported edits remain an explicit native step.

The same requirement applies to native field coverage, reply events and recurrence changes whether n8n or the custom service invokes the vendor. Neither platform supplies an unavailable vendor operation.

# 04 Invoicing payroll and accounting

Start by improving native setup and reviewed exceptions around the existing financial process. The objective is less searching and re-entry, with the bookkeeper retaining financial decisions.

| Opportunity and pain points | Effective treatment | Retained responsibility |
|---|---|---|
| Billing exceptions — PP044–045, PP051–053 | Compare current attendance, notes, service/category and invoice evidence; use agreed cutoffs and mappings. | Bookkeeper decides billable treatment and resolves unusual/rounding cases. The rare export issue is not a first priority. |
| Prepayments and recipients — PP046–048 | Maintain approved instruction references and stable native/Xero IDs; expose verified changes, credits and recipient exceptions. | Xero owns the ledger. No parallel editable credit balance or automatic tax/credit decision is introduced. |
| Returned/overdue invoices — PP049–050, PP054 | Link rejection reason, external invoice, correction history and owned follow-up. Add authorized Xero status reads/events where useful. | Bookkeeper approves correction/reissue and verifies payment. Sending is not acceptance or payment. |
| Pay-run preparation — PP055–060 | Check native staff/pay-item links, periods and run population; compare approved source totals; flag missing timesheets. | Confirm the payroll vendor and calendar. Missing data is not zero pay; leavers may still require final pay. |
| Adjustments and units — PP061–065 | Retain original/approved values, period, reason, approver and separate hours/km/currency. | Payroll approves leave, back pay, applicable rules and first-job travel. Garbled transcript figures are not configuration. |
| Finalization — PP066 | Use existing payroll and accounting approvals/exports; record the checked handoff. | Payroll finalization, payment release and payslip delivery remain in the authorized payroll process. |

ShiftCare's documented integration can export invoices to Xero drafts and export timesheets after staff/pay-item setup; contractor timesheets are excluded and pay periods must match. Validate OCD's configuration before rebuilding those transfers. ShiftCare integration guide

The repository has no direct Xero connector implementing OAuth, status reconciliation or payment events. Those capabilities are planned work if needed. The custom system can implement them directly; n8n's Xero connector could reduce transport setup but does not replace identity links, permissions, reconciliation or approved treatment.

n8n's documented Xero node covers contact and invoice operations; unsupported operations require HTTP requests or custom work. Its presence does not establish complete payroll integration. Xero node

# 05 People processes and organizational constraints

Many items can be made more visible without turning them into new software modules. Their primary remedy is operational ownership.

| Pain points | Suitable support inside the agreed scope | Work that remains outside automated judgement |
|---|---|---|
| Procedures and training — PP074–079 | Current procedure links, short task checklists, source timestamps and observed-practice references. | Trainer assesses understanding and telephone readiness. No new LMS or automated competence scoring. |
| Access and tool setup — PP080–084 | Assigned onboarding/setup checklist and verification record in existing administrative work. | Account administrators configure permissions, phone routing, groups, audio and meeting access. |
| Offboarding — PP086 | Coordinate website, payroll and access steps with distinct owners and checked results. | Authorized administrators perform their native changes; records are not deleted indiscriminately. |
| Owner dependence — PP087–088, PP095 | Owned cases, handover view, deadlines, delegated decision rules and approved closure dates. | Management provides actual coverage and authority. A checklist alone does not make the business independent. |
| Staffing and revenue — PP089–091 | Native vacancy/readiness evidence and basic actual-outcome/cost measures. | Recruit, assess, train and fund workers. Neither custom code nor n8n creates missing service capacity or cash. |
| Marketing and supplier administration — PP092–094 | Retain agreed tasks/reminders in existing tools; reuse current operational reports where relevant. | No new campaign engine, migration product or forecasting model is included. |
| Duplicate work — PP096 | Native-first feature review, stable IDs and one authoritative write path per operation. | Owners decide which remaining gaps justify custom functionality. |

Some small reminders or approved publishing actions might be convenient in n8n if OCD already operates it. These peripheral uses do not justify making the core onboarding, roster or finance lifecycle depend on a second platform.

Questions about the report fragment, employee-contract fragment and certificate task remain clarification work. The seven clarification themes in the pain-point inventory are retained as questions; this assessment does not manufacture new requirements from them.

# 06 Objective comparison of the approaches

Both approaches can execute API calls, schedules, conditions and human-review steps. The distinction is how much domain behavior and operational responsibility the team must maintain around them.

| Criterion | Custom system owns automation | n8n is the primary engine |
|---|---|---|
| Initial delivery | More execution/connector plumbing; existing custom runtime reduces new work. | Built-in nodes and visual flows speed common integrations; OCD-specific cases and rules still need design. |
| Staff experience | One purpose-built review interface with evidence, ownership and business status. | Native n8n forms/waits can support review; a sustained case workspace and domain access may still need custom work. |
| Business rules | Typed services, shared validation, migrations and direct unit/integration tests. | Conditions/code nodes/subflows are flexible; shared domain logic needs disciplined reuse and test coverage. |
| Failure handling | Team owns durable dispatch, retries, reconciliation and operator tools. | Execution tooling/retries are provided; ambiguous vendor writes still need application-level reconciliation. |
| Integrations | Team maintains authentication, paging, mapping, quotas and API changes. | Nodes can reduce this burden; HTTP/custom nodes still need vendor-contract validation. |
| Audit and access | Case-specific history, approvals and branch access can be designed with the business model. | Execution history and platform access are useful; they are not automatically OCD's case retention or per-record authorization model. |
| Maintainability | Familiar repository/CI, but risk of overengineering and dependence on one developer. | Visual accessibility and reusable subflows, but risk of sprawling graphs, embedded code and two logic owners. |
| Portability and scope | Own source, schema and lifecycle; infrastructure/libraries still require upkeep. | Workflow definitions export, but node semantics and runtime upgrades remain n8n-specific. |

n8n genuinely supports durable waits: it can persist a paused execution and resume on a webhook or form submission. Therefore the case for custom ownership is not that n8n cannot do approvals. It is the combination of OCD's review model, existing implementation, stable records and tightly coupled checks. Wait node

Either choice can be reliable or poorly maintained. Source quality, validated vendor operations, regression testing and a funded owner determine the practical result.

# 07 Scalability and maintainability

**Custom path:** Start with the existing web/API application, a separately running worker, managed PostgreSQL and protected file storage. Scale API and workers independently after measuring queue age, latency and API budgets. Keep a modular application codebase; future branches do not initially require separate microservices or separate installations.

Use a mature durable job component rather than writing all broker semantics from scratch. A PostgreSQL-backed option such as pg-boss uses concurrent row claiming and supports background execution while limiting additional infrastructure. Validate runtime compatibility and outage behavior before adopting it. A broker does not make external side effects exactly once. pg-boss project

**n8n path:** Queue mode scales execution across workers using Redis and a shared database. Distributed queue operation does not support SQLite, and filesystem binary storage is incompatible with queue mode. That architecture is viable, but it adds services and operational coordination compared with a PostgreSQL-backed custom worker. Hosted plan limits must be checked independently. Queue mode

**Shared limits:** More workers cannot create API quota or native feature access. Use a limiter per vendor account/organisation, incremental reads, bounded backfill, fair scheduling and replay-safe input handling. An OCR backlog should not block urgent cancellation review. Scale from measured load; 96 pain-point labels are not a volume estimate.

**Future branches:** Add explicit branch/account/organisation scope to cases, links, schedules, files and authorization. Separate branch visibility from accounting-organisation boundaries. Reuse configured workflows instead of copying a complete workflow for every branch. Test cross-branch denial, not only successful access.

**Change management:** Both paths need separate test/production configuration, staged changes and restore practice. n8n's built-in Git-backed environments are Business/Enterprise features; JSON exports can still be versioned outside that feature. Custom Git/CI also needs actual review discipline. Source control

# 08 Recommended custom architecture

**One modular supporting application with an independently operated worker is sufficient for the current scope.** Keep business state separate from background job attempts.

| Component | Required responsibility |
|---|---|
| Existing React office workspace | Show evidence, original sources, matched identities, exact proposed changes, owners and unresolved handoffs. Reuse native worker functions. |
| Existing Hono/TypeScript API | Authenticate staff; enforce account/branch/case permissions; validate requests and maintain business transitions. |
| Workflow/domain services | Reuse W01–W14 preparation, approval and verification; keep shared identity, policy, matching and delivery rules in one place. |
| PostgreSQL business records | Separate cases, source versions, approvals, actions/attempts, history, entity links and connector state into queryable records with appropriate constraints. |
| Durable jobs and worker | Process extraction, reads, approved dispatch and verification asynchronously, with leases/heartbeat, bounded retries and operator-visible failures. |
| Scheduler and outbox | Persist due slots and catch-up policy; couple business events and queued work transactionally or through a durable outbox relay. |
| Protected evidence storage | Retain originals, hashes, allowed access and retention metadata. Object storage is appropriate as files grow; pass protected references to jobs. |
| Vendor adapters | Own ShiftCare operations, Graph reads, necessary Xero reads/events and the approved delivery/signing handoffs. Keep secrets server-side. |
| Operations and history | Monitor worker health, source freshness, queue age, reconciliation, incomplete handoffs and delivery failures; maintain recoverable backups. |

A PostgreSQL job library is a component of the custom application, not a second business platform. Its operational queue tables must not replace the durable case/approval history.

Keep a narrow interface between preparation and external effects. Only an authorized domain command can dispatch a reviewed action. UI buttons, incoming events and optional adapters all call the same checked path.

# 09 Execution and integration ownership

**Lifecycle:** capture source → deduplicate/match → prepare → resolve missing evidence → review → approve exact version → queue action → preflight current native state → dispatch → independently verify → complete required handoffs → close.

Persist a source key/version and external account scope. Bind approval to the source, proposed fields and policy version. A later edit invalidates the earlier approval. Before a native write, check current assignments, locks, affected people and financially relevant state.

Record an action/attempt before dispatch. Retry bounded reads and definite transient preparation failures. After a write timeout, an operator-visible unknown outcome must be reconciled against native records before another potentially duplicate create/cancel/export. There is no single transaction spanning PostgreSQL, ShiftCare and Xero.

| Integration | Ownership and constraint |
|---|---|
| ShiftCare | Custom adapter uses validated machine operations; retain supported native UI handoffs for unsupported fields, groups or recurrence changes. Confirm the actual account entitlement. |
| Microsoft 365 | Reuse Graph token acquisition, per-folder delta cursors and immutable source IDs; handle paging, expired cursors, throttling and agreed backfill. Outgoing mail requires its own approved design. |
| Xero | Keep the native invoice/timesheet export path. Add a custom OAuth/token client and ID-linked status reconciliation only for a demonstrated gap. Signed webhook events require replay protection and periodic reconciliation. |
| Signing and delivery | Preserve existing providers and communication policy. Record authorization, dispatch, provider result and required response separately. Unsupported delivery remains an owned human step. |

Graph documents incremental message tracking directly; n8n is not needed to obtain this capability. Graph delta queries

ShiftCare's API-key guide and pricing describe access differently. Confirm entitlement and exact operation/field coverage with the real account before promising automation. n8n and custom code face the same restriction. API-key guide, pricing

AI may assist extraction or drafting, with provenance and review. Keep money, eligibility and care decisions in approved deterministic rules and human review. Xero prohibits training/fine-tuning AI models on its API data; any third-party data processing also needs appropriate consent and vendor-term review. Xero developer terms

# 10 Existing implementation and material gaps

The repository already contains useful custom automation; handler coverage is not proof that every production step works. A fresh backend build and 47 isolated automation tests passed on 24 October. Those tests use fictional evidence/mocks, not OCD production acceptance.

| Finding from current code | Practical consequence / work needed |
|---|---|
| Hono API and React workspace exist. Several older documents still say NestJS. | Extend the current code; treat the framework labels in older prose as stale. |
| Service implements source deduplication, proposal hashes, leases, persisted dispatch and reconciliation states. | Preserve and test these controls rather than splitting their logic into n8n. |
| PostgreSQL stores the whole account state in one JSON row locked for mutation. | Move growing cases/actions/history to separate rows with indexes and uniqueness constraints before higher concurrent branch traffic. |
| Worker prepares cases; execution is exposed through a staff API action. | A general durable approved-action worker with queued verification/retry policy is additional production work. |
| Schedules use one account timezone and current-day slots; same-workflow configuration is singular. | Define branch scope, approved periods, DST where relevant and missed-run catch-up. Avoid silent skipped cutoffs. |
| W07 invokes native cancellation without provider-travel/mileage cleanup. | Add verified cleanup/preconditions or an explicit native handoff before calling the entire case complete. |
| Direct Xero integration and automatic communication delivery are absent. | Keep native financial exports; build necessary status/delivery adapters only after scope and permissions are agreed. |
| Collectors/imports have deliberate batch/file limits; evidence is kept in account state and database/file/Redis storage. | Add measured partitioned collection, storage/retention management and complete-scope reporting for larger runs. |
| Roles/account scope exist; case schema has no production branch authorization model. | Add scoped permission checks for case/file/action access before extending to branches. |
| n8n W03 draft is inactive and lists missing source, matching, approval, native operation and durable storage connections. | These artifacts are an example, not evidence that OCD requires n8n in production. |

Key code: `backend/application.ts`; `backend/automation/service.ts`; `store.ts`; `handlers.ts`; `shiftcare-operations.ts`; `microsoft365.ts`; `scripts/automation-worker.cjs`; `workflows/n8n/w03-review.json`.

# 11 Long term cost comparison

Cost depends on actual source volume, attachments, polling frequency, active workflows, support needs and recovery work. There is no measured workload, maintenance budget or accepted ROI baseline in the supplied requirements, so a defensible currency estimate needs those inputs.

| Cost category | Custom primary | n8n primary |
|---|---|---|
| Shared costs | Native subscriptions/API entitlement, mappings, policy decisions, cleanup, staff review, acceptance and training. | The same costs remain; a workflow node does not remove them. |
| Initial implementation | Domain UI, job integration, adapters and operational controls; reuse current code. | Workflow design plus any custom UI/domain services, nodes and contract validation. |
| Recurring platform | API/worker hosting, PostgreSQL, storage, monitoring and backups. | Cloud or paid self-hosted plan where needed; self-hosted infrastructure, queue services and storage as applicable. |
| Maintenance | Runtime/dependency updates, API changes, regressions, migrations, incidents and documentation. | n8n/node upgrades, workflow regressions, credentials, custom extensions and operational support. |
| Growth driver | Case/history/file volume, job throughput, API budgets and engineer time. | Execution volume/concurrency and plan features, plus the same source/API and support costs. |

**36-month model:** shared costs + initial implementation + 36 × recurring platform/support cost + planned change and incident allowance. Include staff review and recovery time in each option; avoid double-counting shared work or n8n Cloud's included hosting.

n8n pricing is based on workflow executions rather than node count. Production governance and scaling features vary by plan. Its Community license supports internal business use, but some enterprise code/features have separate licensing. Self-hosting removes neither infrastructure nor operator cost. Pricing, license

Custom is favored here by reuse and the concentration of domain-specific work, not a guarantee of lower total cost. n8n can be cheaper for connector-heavy, simpler processes maintained by a capable workflow team. Buy or add it only when a scoped comparison demonstrates a net reduction in delivery and support effort.

# 12 Delivery plan and acceptance

| Stage | Scope | Acceptance before expansion |
|---|---|---|
| 0 — Confirm and configure | Actual API account; native Xero integration; payroll vendor; forms, mappings, rules, source scopes, roles and maintainer. | Office/bookkeeper sign off the applicable process. Baseline handling and errors are recorded. |
| 1 — Shared foundation and input | Queryable cases/jobs/history; access; recovery; Graph intake; reviewed onboarding and document evidence. | Duplicate/changed sources, ambiguous matches, unavailable extraction and worker restart recover correctly. |
| 2 — Operational follow-through | Confirmation replies, reviewed cancellations/cover, instructions, renewal and factual care exceptions. | Travel cleanup, occurrence scope, native locks, no cover, failed delivery and unknown write outcomes stay correctly owned. |
| 3 — Finance reconciliation | Approved period reads, necessary Xero status connector, recipient/prepayment exceptions and payroll preparation handoffs. | Native export has a single owner; totals, identities and current period reconcile. No duplicate invoice/pay release. |
| 4 — Branches and optional arrival | Branch/account permissions, configured schedules and partitioned workload; separately accepted W14 where needed. | Cross-branch denial, missed-run catch-up, rate-limit isolation and real-device/freshness behavior pass. |

These are dependency-based stages, not a promised schedule. Cheap data/configuration fixes can start early; high-risk mutations wait for their policies and verification path.

## Completion scenarios to test

- Same message twice; changed source after approval; two staff acting together; worker dies mid-job.
- Vendor accepts a write but the response is lost; record changes externally before execution; current record cannot be read.
- Duplicate person; incorrect payer; group/locked shift; no travel; unknown mileage rule; incomplete paginated result.
- Expired/revoked token; missing attachment; unsupported/large file; throttling; missed scheduled cutoff; failed recipient delivery.
- Branch/account mismatch; confidential note/file access; audit retention and restoration from backup.

Measure complete-case handling time, unanswered/overdue work, corrections, reconciliation backlog, failed delivery and operator recovery time across comparable periods. Include unsuccessful cases. Do not equate a closed review with completed care, invoice payment or released payroll.

# 13 Responsibilities for optional n8n

**Core workflows must continue to operate with n8n absent.** There is no requirement in the matrix that inherently needs n8n. Optional use needs a specific benefit, not merely technical possibility.

| Responsibility | Decision for this project |
|---|---|
| Cases, source identity, policies, permissions and approvals | Custom system owns the authoritative records and checks. |
| Native onboarding, cancellation, changes, filing and financial decisions | Custom domain service and authorized staff own the approved action and verification. Do not create a competing write path. |
| Schedules, durable jobs and reconciliation | Custom runtime owns the core lifecycle. Do not have n8n and the current worker independently dispatch the same action. |
| Existing Microsoft 365 source | Reuse the custom collector initially. Replacing a working collector needs a measured maintenance advantage. |
| New approved peripheral source | n8n may translate an event into a versioned application ingestion API with a stable source key. |
| Approved non-critical delivery/reminder | n8n may act as an adapter using an application-issued action ID; return a result through an authenticated endpoint. Application owns deduplication and outcome interpretation. |
| Marketing/supplier tooling | Existing administrative tools first. Optional future n8n work needs separate scope and actual value; it is not part of the initial core. |

If used, n8n accesses the minimum necessary fields/references through a narrow API. It must not write directly to domain tables, bypass review, decide a financial treatment or report a native action complete merely because a workflow ran.

n8n's paused executions are durable, but finished execution data can be pruned. Keep business approvals, required evidence and final outcomes under OCD's own retention model. Wait behavior, execution data

Reconsider primary n8n only if measured connector maintenance dominates, governance fits the selected edition and the team can preserve a single domain owner. Reconsider custom scope if OCD cannot fund engineering ownership; start with native/configuration improvements and a smaller supported pilot.

# 14 Existing research and verification basis

## Requirements and prior work reused

- `docs/pain-points-2026-10-23/pain-points.json`: 96 numbered issues, source keys and seven clarification themes. Every ID maps to exactly one row in the opportunity matrix.
- `docs/OCD_Client_Needs_and_Automation_Requirements.md` and the 27-item register: client requests, constraints and proposals kept distinct.
- `docs/OCD_Workflow_Use_Cases.md`: established W01–W14, native ownership, human handoffs and measurement plan.
- `docs/OCD_ShiftCare_Integration_and_Automation_Design.md`: validated-method boundaries, retry/reconciliation and existing financial path.
- `docs/OCD_Native_Automation_Implementation.md`: implementation narrative, checked against actual source because some framework/status labels are stale.
- Supplied CSV coverage and 14/21 October analyses: source-backed manual work; ambiguous payroll/date/contract/report wording remains unresolved.

## Existing diagrams reused as conceptual references

- All 14 workflow connections: preserve the same workflow families and dependencies.
- Integration and data ownership: preserve native sources of truth and owned handoffs.
- Execution and recovery: preserve source matching, exact approval, dispatch, independent verification and reconciliation.

The candidate specifications and connections were inspected. They are dated proposal/prototype diagrams, not current deployment proof. In particular, the integration diagram's GET-only connector label predates opt-in adapters in code. This assessment does not silently treat either label as proof of OCD's live entitlement.

**Matrix source keys:** C05 = 5 October requirements register; R14 = 14 October meeting review; R21 = recording analysis prepared 21 October; C20 = the supplied two-CSV coverage. Their detailed timestamps/lines remain in the original pain-point source register.

## Fresh local checks

Backend build and `tests/native-automation.test.cjs` plus `tests/native-completion.test.cjs`: **47 passed, zero failed/skipped**, with live connector credentials disabled and fictional/mocked operations. No live account actions were performed. These tests establish selected local behavior, not production load, security acceptance, extraction accuracy or successful live vendor integration.

# 15 Current official integration references

Official sources were checked on 24 October 2026. Runtime entitlements, current API headers and actual account behavior still control implementation.

- ShiftCare Xero setup and exports: reuse native drafts and eligible timesheet transfer.
- ShiftCare API keys and pricing: confirm account access where published entitlement descriptions differ.
- Xero rate limits and developer pricing: use current tier/response headers rather than a universal assumed daily quota.
- Xero idempotent requests: use supported keys for eligible requests; still reconcile side effects and other vendors.
- Xero developer terms: data-use restrictions apply to either architecture.
- Microsoft Graph message delta: custom incremental collection can use vendor-supported cursors.
- pg-boss: a candidate PostgreSQL-backed job component, not an installed dependency or accepted implementation.
- n8n Wait node and Xero node: durable waits and bounded built-in operations.
- n8n queue mode: worker/broker/database and binary-storage considerations.
- n8n source control, execution retention, pricing and license: account for edition, execution volume, retention and operational ownership.

## Matrix interpretation

The appendix reproduces each existing pain point and its proposed treatment. The CSV/JSON additionally retain sources, completion criteria, delivery stage and optional n8n role. Treatments and stages are recommendations, not client-approved requirements. An A/R entry can still depend on a manual external handoff when the required interface is unavailable.

No new care system, ledger, payroll calculation engine, hiring decision engine, campaign platform or predictive model is introduced. Hardware fixes, service capacity and human judgement are explicitly retained where software cannot resolve the underlying problem.

# Appendix 01 Opportunities PP-001 to PP-006

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-001 Repeated data entry.** Staff copy information from emailed forms and scanned assessments into participant and worker records in the relevant systems.<br>C05:A01; R14:M1-01; R14:M1-02 | Import original forms, extract draft fields, validate required values and match existing identities before native entry.<br>W03/W04 · R · Stage 1<br>n8n: None required | **Review:** Office or HR approves identity and changed fields.<br>**Evidence:** Original source, approved fields and verified native ID are linked. |
| **PP-002 Scattered referral channels.** Website, email, telephone, coordinator and provider referrals require manual collection and processing.<br>C05:A01 | Capture authorized email, form and staff-logged phone referrals into one deduplicated owned case.<br>W01/W02 · A · Stage 1<br>n8n: Optional source adapter only | **Review:** Office confirms uncertain referral identity.<br>**Evidence:** One source produces one owned enquiry with a next action. |
| **PP-003 Late serviceability checks.** People can complete intake paperwork before learning that OCD cannot serve their area.<br>C05:A02 | Check the approved service area before the full intake; route uncertain areas to staff review.<br>W01 · A · Stage 1<br>n8n: None required | **Review:** Coordinator confirms actual capacity.<br>**Evidence:** No geographic result is presented as a reserved service. |
| **PP-004 Incomplete intake information.** Missing dates create clarification and follow-up work. The intended date field still needs confirmation.<br>C20:TR-31 | Define the intended date field, validate it and assign clarification for missing or ambiguous dates.<br>W01/W03 · C · Stage 0<br>n8n: None required | **Review:** Intake owner approves the form definition.<br>**Evidence:** Required date meaning and source are recorded. |
| **PP-005 Unclear client stages.** Staff search prospect and active-client lists; a consultation record was reportedly moved to active status prematurely.<br>R21:NEW-06 | Use explicit enquiry, consultation and active stages with transition checks against the native record.<br>W01/W03 · R · Stage 1<br>n8n: None required | **Review:** Office approves activation.<br>**Evidence:** Consultation cannot silently become an active service. |
| **PP-006 Finding agreed service hours.** Approved hours and service changes must be recovered from emails and notes, with uncertainty about whether a colleague recorded the update.<br>R14:M1-04 | Link approved service hours and effective changes to their source and affected bookings.<br>W08/W13 · R · Stage 2<br>n8n: None required | **Review:** Coordinator approves the allocation change.<br>**Evidence:** Saved schedule agrees with the approved instruction. |

# Appendix 02 Opportunities PP-007 to PP-012

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-007 Duplicated inbox attention.** Multiple administrators inspect the same messages while other enquiries remain unanswered. C05:A03 | Deduplicate incoming messages and show assigned owner, due time and response status. W01/W02 · A · Stage 1 n8n: None required | **Review:** Office resolves conflicting ownership. **Evidence:** One actionable enquiry has one accountable owner. |
| **PP-008 Manual channel monitoring.** Staff repeatedly check Outlook, Dialpad, messages and voicemail while performing other work. R14:M1-26; R21:NEW-07 | Use the existing Graph delta collector for authorized email and owned manual references for unsupported phone events. W02/W13 · A · Stage 1 n8n: Optional connector for an approved additional channel | **Review:** Office confirms available call-event access. **Evidence:** New work is visible without treating unread status as ownership. |
| **PP-009 Repetitive correspondence.** Similar emails and updates are prepared participant by participant. R14:M1-03 | Prepare messages from reviewed templates and an approved delivery outbox. W13 · R · Stage 3 n8n: Optional delivery adapter behind the application API | **Review:** Office approves recipient and content. **Evidence:** Delivery result or owned failure is retained. |
| **PP-010 Incomplete messages.** Messages can omit the person, service time, confirmed outcome or next action. C20:TR-24; R21:NEW-18 | Validate person, date/time, confirmed change and next action before a message is approved. W13 · A · Stage 1 n8n: None required | **Review:** Office checks the actual message. **Evidence:** Required context is present before sending. |
| **PP-011 Unverified communication outcomes.** Staff separately check whether people were notified, received the message or responded. R14:M1-09; C20:TR-23; C05:A11 | Track approval, dispatch, provider receipt, response and failed delivery as separate states. W13 · R · Stage 3 n8n: Optional delivery adapter only | **Review:** Office handles uncertain receipt or no response. **Evidence:** A sent action is never treated as acknowledged without evidence. |
| **PP-012 Scattered communication evidence.** Establishing what was communicated requires searching across channels, while protecting access to confidential information. C05:A11; R14:M1-20 | Index authorized communication references against the person and case with role-based access. W13/Shared · A · Stage 1 n8n: None required | **Review:** Office controls disclosure and history scope. **Evidence:** Reviewer can locate the relevant communication without cross-role leakage. |

# Appendix 03 Opportunities PP-013 to PP-018

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-013 Manual prioritization.** Staff identify urgent requests, connect complaints to the relevant invoice or booking, and decide who should act. C05:A03 | Suggest category and urgency and retrieve candidate booking/invoice references for review. W02/W10 · R · Stage 1 n8n: None required | **Review:** Office verifies identity, urgency and the relevant record. **Evidence:** Uncertain matches remain assigned exceptions. |
| **PP-014 Arrival enquiries.** Participants call the office to ask whether a worker is on the way or why the worker has not arrived, adding interruptions. C05:A12 | Evaluate native arrival visibility first; if a gap remains, expose a consented, fresh ETA for the assigned service. W14 · R · Stage 4 n8n: None required | **Review:** Office and worker confirm sharing scope. **Evidence:** Stale, reassigned or unauthorized estimates are withdrawn. |
| **PP-015 Complex service capacity.** Coverage depends on location, worker hours, departures, preferences and actual capacity, rather than postcode alone. C05:A02; C05:A05 | Combine configured area checks with a coordinator capacity decision and current roster evidence. W01/W08 · R · Stage 1 n8n: None required | **Review:** Coordinator decides capacity and negotiated exceptions. **Evidence:** Coverage information does not overpromise available workers. |
| **PP-016 Uncertain worker availability.** An empty calendar slot does not establish willingness or availability to accept work; worker confirmation is still required. C05:A05 | Check roster/leave and record worker acceptance before confirming a proposed booking change. W08 · R · Stage 2 n8n: None required | **Review:** Worker and participant confirm the arrangement. **Evidence:** An unbooked slot alone cannot confirm work. |
| **PP-017 Manual booking creation.** Staff copy existing bookings and change dates, times, workers, service details and travel information. R21:NEW-05 | Prepare a booking draft from reviewed call details and validate dates, service and overlaps. W08 · R · Stage 2 n8n: None required | **Review:** Coordinator performs unsupported creation in ShiftCare. **Evidence:** One correctly scoped native booking is read back. |
| **PP-018 Recurring change ambiguity.** Staff clarify whether a change affects one occurrence or future bookings before editing. R14:M1-05 | Preview one-occurrence versus future-series effects and retain the approved scope. W08 · R · Stage 2 n8n: None required | **Review:** Coordinator authorizes the recurrence scope. **Evidence:** Native changes match the approved occurrence set. |

# Appendix 04 Opportunities PP-019 to PP-024

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-019 Time entry errors.** AM/PM and service-time misunderstandings require clarification and correction. R14:M1-08 | Use explicit date, timezone and 24-hour previews before saving reviewed times. W08 · A · Stage 2 n8n: None required | **Review:** Coordinator checks exceptional times. **Evidence:** The saved occurrence has the intended timestamp. |
| **PP-020 Repeated save verification.** Staff reopen records and future occurrences to check that changes saved correctly. The cause of an apparent failed update was not established. R14:M1-08 | Retrieve the saved native record and compare expected fields before reporting completion. W07/W08 · A · Stage 2 n8n: None required | **Review:** Office reconciles mismatched or unknown results. **Evidence:** Completion requires matching native evidence. |
| **PP-021 Manual travel checks.** Addresses, kilometres, travel duration and separate travel/service blocks require checking. R14:M1-06; R14:M1-07; R21:NEW-03 | Calculate proposed route estimates and compare travel/service blocks using confirmed addresses. W08/W11 · R · Stage 2 n8n: None required | **Review:** Coordinator or payroll reviewer approves eligible travel treatment. **Evidence:** Route estimates remain distinct from actual mileage and pay. |
| **PP-022 Stale travel information.** Worker or booking changes can leave previous travel values in place. R21:NEW-03 | Invalidate or flag saved travel when its worker, address or preceding booking changes. W08 · A · Stage 2 n8n: None required | **Review:** Coordinator reviews the changed travel context. **Evidence:** A stale estimate cannot remain silently approved. |
| **PP-023 Recurring reassignment work.** A departing worker's bookings require reassignment, travel review and participant communication, even where future-shift updates are available. R21:NEW-04 | Coordinate native future-shift reassignment, travel review and affected-participant notices as one case. W08/W13 · R · Stage 2 n8n: None required | **Review:** Coordinator selects cover and approves scope. **Evidence:** Every affected native occurrence and required notice is checked. |
| **PP-024 Unconfirmed service times.** Staff chase participants or families to establish when workers should attend. R14:M1-10 | Create deadline-driven clarification tasks for unconfirmed service times. W08/W13 · A · Stage 2 n8n: Optional reminder delivery only | **Review:** Office obtains the participant or family decision. **Evidence:** Final agreed time is recorded before confirmation. |

# Appendix 05 Opportunities PP-025 to PP-030

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-025 Repeated instruction maintenance.** Staff inspect bookings individually and copy missing shift or timesheet instructions.<br>R21:NEW-01 | Identify missing instructions and preview application of an approved instruction template to selected bookings.<br>W08/Shared · R · Stage 2<br>n8n: None required | **Review:** Coordinator approves wording and affected scope.<br>**Evidence:** Correct instructions are verified on the intended bookings. |
| **PP-026 Instruction version confusion.** Instructions are rewritten and formatted in separate tools, with manual before/after tracking.<br>R21:NEW-02 | Keep reviewed instruction versions, effective dates and change history with structured fields.<br>Shared/W08 · R · Stage 1<br>n8n: None required | **Review:** Procedure owner approves the current version.<br>**Evidence:** Staff can identify which wording applies to a booking. |
| **PP-027 Daily confirmation checking.** Staff manually inspect reminder responses and determine the required action.<br>C05:A04 | Match native reminder replies to the exact occurrence and route yes/no/change exceptions.<br>W06 · R · Stage 2<br>n8n: None required | **Review:** Office resolves ambiguous identity or occurrence.<br>**Evidence:** Silence and ambiguous replies never cancel a service. |
| **PP-028 Incomplete cancellations.** A worker can be notified while the corresponding booking remains active.<br>C20:TR-23; R21:NEW-10 | Require verified booking change plus financial and communication handoffs before case closure.<br>W07/W13 · R · Stage 2<br>n8n: None required | **Review:** Office checks unresolved native or contact steps.<br>**Evidence:** Notification alone cannot close the cancellation. |
| **PP-029 Cancellation travel corrections.** Provider travel and mileage require separate review and cleanup when travel did not occur.<br>C20:TR-19; R21:NEW-11 | Add explicit actual-travel evidence and a checked travel/mileage cleanup step before cancellation where policy requires it.<br>W07 · R · Stage 2<br>n8n: None required | **Review:** Bookkeeper approves treatment; staff perform unsupported native edits.<br>**Evidence:** Travel lines and cancellation status both match the approved result. |
| **PP-030 Different cancellation treatments.** Staff distinguish participant cancellations from provider inability to deliver, then establish the approved billing treatment.<br>R14:M1-16; C05:A04 | Version the approved cancellation cause, notice and charge rules before enabling a supported action.<br>W07 · C · Stage 0<br>n8n: None required | **Review:** Bookkeeper or authorized manager approves the policy.<br>**Evidence:** Exact occurrence and money consequences are approved together. |

# Appendix 06 Opportunities PP-031 to PP-036

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-031 Last minute absences.** Sickness and resignations trigger urgent coordination and roster changes.<br>C05:A06 | Create a disruption case listing affected bookings and time-bound cover tasks.<br>W08 · A · Stage 2<br>n8n: Optional reminder delivery only | **Review:** Coordinator obtains available-worker and participant decisions.<br>**Evidence:** Every affected service has a confirmed outcome or owned escalation. |
| **PP-032 Insufficient backup capacity.** Several simultaneous absences can exceed the available replacement pool.<br>R14:M1-13 | Show uncovered services and available/ready-worker evidence using the native roster; track the recruitment dependency.<br>W08/W04 · H · Stage 2<br>n8n: None required | **Review:** Management supplies actual staffing capacity.<br>**Evidence:** Uncovered work stays visible instead of being auto-confirmed. |
| **PP-033 Individual participant decisions.** Each affected participant may prefer replacement cover, rescheduling or skipping the visit, requiring individual confirmation.<br>R14:M1-14; C20:TR-30 | Track replace, reschedule or skip decisions separately for each affected participant.<br>W08/W13 · R · Stage 2<br>n8n: None required | **Review:** Participant and coordinator agree the outcome.<br>**Evidence:** Each participant has an agreed outcome and native record reference. |
| **PP-034 Replacement suitability checks.** Staff consider geography, availability, continuity, introductions and participant agreement.<br>R14:M1-15 | Present eligibility, geography, continuity and acceptance evidence beside native cover options.<br>W08 · R · Stage 2<br>n8n: None required | **Review:** Coordinator selects suitable cover; worker and participant agree.<br>**Evidence:** No autonomous assignment is made from distance alone. |
| **PP-035 Fragmented document storage.** Files must be located across email, cloud storage and ShiftCare.<br>R21:NEW-08 | Index original document locations and link the authoritative native filing result without copying entire care databases.<br>W05/W12 · A · Stage 1<br>n8n: None required | **Review:** Office confirms the correct person and document.<br>**Evidence:** Original and native document references are findable together. |
| **PP-036 Manual filing.** Received documents must be matched and placed under the correct person.<br>C05:A10; R21:NEW-08 | Match person, type, version and visibility, then use a supported upload or verified native filing handoff.<br>W05 · R · Stage 1<br>n8n: None required | **Review:** Office checks signature, person and disclosure.<br>**Evidence:** The intended native record contains the correct file and metadata. |

# Appendix 07 Opportunities PP-037 to PP-042

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-037 Record by record completeness checks.** Staff open individual records to establish whether required paperwork is present. A missing native copy does not prove the agreement does not exist.<br>R21:NEW-08 | Produce a scoped completeness checklist from expected document requirements and available evidence.<br>W05/W12 · R · Stage 1<br>n8n: None required | **Review:** Office checks suspected missing copies against originals.<br>**Evidence:** Missing native copy is not equated with absent agreement. |
| **PP-038 Missing attachments.** Emails can refer to documents that were not attached, creating another follow-up.<br>R21:NEW-21 | Check expected attachments before dispatch and create an owned follow-up for incomplete received requests.<br>W05/W13 · A · Stage 1<br>n8n: Optional delivery adapter only | **Review:** Sender or reviewer resolves the missing file.<br>**Evidence:** The referenced document is actually available to the intended recipient. |
| **PP-039 Unclear document dates.** Staff interpret date formats and distinguish plan dates from agreement dates; an initial expired-plan interpretation was corrected in the discussion.<br>R21:NEW-09 | Extract date candidates with their source text and distinguish plan, agreement and expiry meanings.<br>W05/W12 · C · Stage 0<br>n8n: None required | **Review:** Document owner confirms ambiguous dates.<br>**Evidence:** No expired or non-expiring classification is invented. |
| **PP-040 Coordinator follow-up.** Updated dates, funding information and supporting evidence require external clarification.<br>R21:NEW-09 | Assign requests for updated plan dates or funding with an owner, due date and retained response.<br>W05/W13 · A · Stage 2<br>n8n: Optional reminder delivery only | **Review:** Coordinator or document owner verifies the supplied information.<br>**Evidence:** Verified update is linked to the correct native record. |
| **PP-041 Incorrect expiry alerts.** Non-expiring documents can appear expired because their metadata was entered incorrectly.<br>C05:A07 | Compare original evidence with configured expiry metadata and propose corrections.<br>W12 · R · Stage 2<br>n8n: None required | **Review:** Authorized reviewer confirms real expiry treatment.<br>**Evidence:** Correct native metadata is read back before an alert is resolved. |
| **PP-042 Renewal chasing.** Genuine missing or expiring worker documents require checking and follow-up.<br>C05:A07 | Schedule genuine missing/expiring-document checks and assign renewal work under approved requirements.<br>W12/W13 · A · Stage 2<br>n8n: Optional reminder delivery only | **Review:** HR confirms requirements and worker readiness.<br>**Evidence:** Renewal remains open until acceptable evidence is verified. |

# Appendix 08 Opportunities PP-043 to PP-048

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-043 Policy identification.** Staff locate the correct company policies and distinguish them from agency paperwork.<br>R14:M1-21 | Maintain links to the current approved company policies and required onboarding checklist.<br>Shared/W04 · H · Stage 1<br>n8n: None required | **Review:** HR or procedure owner chooses the correct policy.<br>**Evidence:** Staff receive the applicable version, not substituted agency paperwork. |
| **PP-044 Service by service billing review.** Staff compare completed work, attendance, notes, agreed hours and billing details.<br>C05:A08; C05:A09 | Assemble scheduled/actual time, notes, invoice lines and agreement evidence into specific billing exceptions.<br>W09/W10 · R · Stage 3<br>n8n: None required | **Review:** Bookkeeper decides billable treatment.<br>**Evidence:** The correction or explanation is bound to current source evidence. |
| **PP-045 Incorrect service coding.** Cleaning/support categories and travel items can be mismatched and require correction.<br>C20:TR-01; C20:TR-20; R21:NEW-14 | Validate actual service against an approved category and travel-item mapping before native export.<br>W10 · R · Stage 3<br>n8n: None required | **Review:** Bookkeeper approves mappings and corrections.<br>**Evidence:** The selected service/item matches its approved source. |
| **PP-046 Client instructions held in memory.** Advance payments and special billing arrangements depend on notes and recollection.<br>C20:TR-02; R21:NEW-15 | Expose approved client billing instructions and read verified Xero prepayment/credit references where authorized.<br>W10 · R · Stage 3<br>n8n: None required | **Review:** Bookkeeper allocates credits and decides financial treatment.<br>**Evidence:** No shadow credit ledger or duplicate allocation is created. |
| **PP-047 Manual tax setting checks.** Staff select and verify invoice tax and accounting settings against the approved instructions.<br>C20:TR-04; R21:NEW-15 | Correct native account/tax mappings with accountant-approved settings and surface exceptions.<br>W10 · N · Stage 0<br>n8n: None required | **Review:** Accountant or bookkeeper approves settings.<br>**Evidence:** Native draft/export reflects the approved mapping. |
| **PP-048 Outdated billing recipients.** Plan-manager changes may not be communicated before invoices are issued.<br>C20:TR-05; R21:NEW-17 | Flag payer/contact changes and maintain stable native/Xero ID links with verification dates.<br>W10/W13 · R · Stage 3<br>n8n: None required | **Review:** Bookkeeper verifies the responsible payer.<br>**Evidence:** Unverified recipient changes cannot silently redirect billing. |

# Appendix 09 Opportunities PP-049 to PP-054

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-049 Returned invoice investigation.** Staff locate the rejected invoice, establish the reason and determine the correction.<br>C20:TR-05 | Link the external invoice, rejection reason, native source and correction history in one case.<br>W10 · R · Stage 3<br>n8n: None required | **Review:** Bookkeeper chooses amendment or reissue treatment.<br>**Evidence:** Corrected external reference and checked outcome are retained. |
| **PP-050 Manual invoice chasing.** Calls, messages, resending and payment verification require staff follow-through.<br>C20:TR-25; R21:NEW-17 | Use authorized Xero status reads/events to prepare owned follow-ups and approved contact actions.<br>W10/W13 · R · Stage 3<br>n8n: Optional approved reminder delivery only | **Review:** Accounts owner confirms outcome and contact.<br>**Evidence:** Paid/closed state requires current external evidence. |
| **PP-051 Unclear billing cadence.** Staff repeatedly clarify when completed services should be invoiced; the discussion did not establish one definitive new schedule.<br>R21:NEW-16 | Configure reviewed cutoffs and completed-service review periods only after the billing cadence is agreed.<br>W10 · C · Stage 0<br>n8n: None required | **Review:** Bookkeeper approves calendar and scope.<br>**Evidence:** No unapproved production billing schedule is inferred. |
| **PP-052 Exceptional invoices need senior review.** Unusual amounts or unclear instructions are handed to experienced staff; general escalation criteria remain to be agreed.<br>C20:TR-03 | Define escalation conditions and route unusual invoices with their evidence to the authorized reviewer.<br>W10 · C · Stage 0<br>n8n: None required | **Review:** Billing owner sets thresholds and makes the decision.<br>**Evidence:** Exception has a named reviewer and recorded disposition. |
| **PP-053 Occasional export discrepancies.** Rounding or export issues require correction; the client described these as lower priority.<br>C05:A09 | Keep rare export/rounding discrepancies in the native bookkeeper reconciliation process initially.<br>W10 · H · Stage 3<br>n8n: None required | **Review:** Bookkeeper corrects the actual export discrepancy.<br>**Evidence:** No duplicate export is used to mask an unresolved difference. |
| **PP-054 Payment delays affect payroll planning.** Owners report concern that returned invoices delay the cash needed for wages.<br>R21:NEW-17; C20:TR-25 | Show verified overdue/returned balances and outstanding follow-up; measure correction delays.<br>W10 · H · Stage 3<br>n8n: None required | **Review:** Management handles actual cash commitments and funding decisions.<br>**Evidence:** Reporting does not promise payment timing or payroll liquidity. |

# Appendix 10 Opportunities PP-055 to PP-060

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-055 Manual pay run configuration.** Period dates, payment dates and payment methods require checking. Weekly/fortnightly wording in the transcript remains unclear. C20:TR-06 | Align approved native pay periods, staff links and pay items before duplicating data transfer. W11 · N · Stage 0 n8n: None required | **Review:** Payroll officer confirms the calendar and vendor. **Evidence:** Native draft timesheet export matches the real pay period. |
| **PP-056 Employee list cleanup.** Inactive workers can remain in payroll selection and need review. C20:TR-07 | Compare the run population with current staff status and surface leaver/starter exceptions. W04/W11 · R · Stage 3 n8n: None required | **Review:** Payroll officer approves final or outstanding pay treatment. **Evidence:** A leaver is neither silently paid nor excluded from required final pay. |
| **PP-057 New starter setup.** Adding an employee requires a separate payroll setup process whose complete steps were not supplied. C20:TR-08 | Track employee setup as a separate native payroll handoff with explicit readiness checks. W04/W11 · H · Stage 1 n8n: None required | **Review:** HR/payroll completes authorized setup. **Evidence:** Worker profile creation does not imply payroll readiness. |
| **PP-058 Previous run values.** Staff inspect and refresh information carried over from earlier payslips. The field described as a base rate needs verification. C20:TR-09 | Identify the actual carried-over field and show previous/current quantities for payroll review. W11 · C · Stage 0 n8n: None required | **Review:** Payroll officer verifies the screen and correction procedure. **Evidence:** No inferred reset changes a contractual wage rate. |
| **PP-059 Worker by worker reconciliation.** Staff compare ShiftCare hours and mileage with payroll inputs and manually enter the approved values. C20:TR-10; R21:NEW-12 | Prepare approved hours/mileage exceptions and reuse native timesheet export where eligible. W11 · R · Stage 3 n8n: None required | **Review:** Payroll officer reviews external draft totals. **Evidence:** Actual entry is reconciled without another competing export path. |
| **PP-060 Missing timesheets.** Incomplete source records require investigation before payroll can proceed. C20:TR-11 | Detect missing or inconsistent timesheet sources and assign resolution before payroll review. W09/W11 · A · Stage 3 n8n: None required | **Review:** Coordinator supplies and approves missing source evidence. **Evidence:** Missing data is never treated as zero hours. |

# Appendix 11 Opportunities PP-061 to PP-066

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-061 Adjustment tracking.** Hour changes and exceptions must be found in notes and applied to the correct period.<br>C20:TR-13; R21:NEW-13 | Attach period-specific adjustments, original values, approval and rule references to the review.<br>W11 · R · Stage 3<br>n8n: None required | **Review:** Payroll officer approves applicability and calculation.<br>**Evidence:** Changed sources invalidate the reviewed preparation draft. |
| **PP-062 Leave verification.** Staff check requests, supporting documents, balances and the applicable payroll treatment.<br>C20:TR-14; R14:M1-12 | Surface leave requests, supporting documents and verified native treatment as payroll exceptions.<br>W11/W04 · R · Stage 3<br>n8n: None required | **Review:** Payroll/HR decides entitlement and election under approved rules.<br>**Evidence:** The system does not calculate legal entitlement from transcript wording. |
| **PP-063 Back pay entry.** Corrections require calculations, approval, entry and explanatory notes.<br>C20:TR-16; R21:NEW-13 | Retain the affected period, approved correction, source calculation and payslip note for back pay.<br>W11 · R · Stage 3<br>n8n: None required | **Review:** Payroll officer approves and enters the correction.<br>**Evidence:** External payroll reference and correction history are checked. |
| **PP-064 Mileage interpretation.** Kilometres, reimbursement amounts and payroll fields require careful distinction; the transcript does not establish a reimbursement rate.<br>C20:TR-15 | Enforce separate distance/time/currency fields after reimbursement rules and units are approved.<br>W11 · C · Stage 0<br>n8n: None required | **Review:** Payroll officer confirms units and rate source.<br>**Evidence:** Kilometres cannot be silently interpreted as dollars. |
| **PP-065 Unconfirmed travel treatment.** Home-to-first-job changes need an approved decision and communication before application.<br>C20:TR-12 | Hold first-job travel changes pending approved treatment, effective date and required communication.<br>W08/W11 · C · Stage 0<br>n8n: None required | **Review:** Authorized manager/payroll approves the rule.<br>**Evidence:** A pending policy discussion cannot change payments. |
| **PP-066 Manual payroll completion.** Staff save, finalize and email payslips employee by employee in the external payroll process.<br>C20:TR-17 | Use the established payroll system for pay-run finalization and payslip delivery; track only the checked handoff.<br>W11 · N · Stage 3<br>n8n: None required | **Review:** Authorized payroll officer releases the run.<br>**Evidence:** A local worksheet is not treated as payment or payslip delivery. |

# Appendix 12 Opportunities PP-067 to PP-072

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-067 Reported incomplete work.** Owners describe missed tasks, inconsistent cleaning and early departures requiring investigation; these are reported concerns about service delivery. R14:M1-17 | Present factual task-completion, attendance and feedback exceptions for supervisor review. W09 · R · Stage 2 n8n: None required | **Review:** Supervisor investigates service concerns. **Evidence:** No worker performance conclusion is inferred automatically. |
| **PP-068 Important information buried in notes.** Significant events and extra time must be found among routine visit records, including restricted office notes. C05:A08 | Flag source-defined significant-note and duration exceptions with restricted access and original links. W09 · R · Stage 2 n8n: None required | **Review:** Responsible staff assess significance and existing escalation. **Evidence:** Summary does not replace care records or urgent incident response. |
| **PP-069 Manual participant feedback.** Staff arrange calls to learn how services are going. R14:M1-18 | Schedule agreed feedback tasks and retain participant responses and resulting actions. W09/W13 · A · Stage 2 n8n: Optional approved reminder delivery only | **Review:** Office conducts appropriate personal contact. **Evidence:** Feedback has an owner and unresolved concerns remain visible. |
| **PP-070 Reactive problem discovery.** The supervisor wants proactive feedback so concerns can be raised before relying on incoming complaints. R14:M1-18 | Use the same agreed feedback follow-up process to help staff raise concerns earlier. W09/W13 · H · Stage 2 n8n: None required | **Review:** Supervisor decides service-quality action. **Evidence:** No automated satisfaction or clinical judgement closes a concern. |
| **PP-071 Fragmented complaint histories.** Relevant explanations and correspondence must be assembled from several sources. R14:M1-19 | Link restricted complaint evidence, communications, deadlines and decisions in an owned case. W13/Shared · R · Stage 2 n8n: None required | **Review:** Manager assesses the complaint and response. **Evidence:** Case access and resolution evidence match the approved process. |
| **PP-072 Pending personnel cases.** Calls, voicemail, replies, management decisions and replacement arrangements must be coordinated. R14:M1-20 | Track pending responses, callbacks and coverage dependencies without making personnel decisions. W08/W13 · H · Stage 2 n8n: None required | **Review:** Management retains the employment decision. **Evidence:** Personnel case and service cover have separately verified outcomes. |

# Appendix 13 Opportunities PP-073 to PP-078

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-073 Safety follow-up.** Reported household hazards require reviewed communication and coordinated action with the appropriate people.<br>R21:NEW-22 | Capture factual hazard reports and assign follow-up through existing incident procedures.<br>W09/W13 · H · Stage 2<br>n8n: None required | **Review:** Responsible manager handles immediate safety and the response.<br>**Evidence:** Software task status cannot replace urgent safety escalation. |
| **PP-074 Scattered procedural knowledge.** Staff rely on personal notes and long recordings to recover instructions.<br>R14:M1-24; C20:TR-21; R21:NEW-19 | Publish approved procedure versions and source timestamps within existing documentation links.<br>Shared/W04 · H · Stage 1<br>n8n: None required | **Review:** Procedure owner maintains accuracy.<br>**Evidence:** Staff can retrieve the current instructions for the task. |
| **PP-075 Repeated explanations.** Supervisors revisit procedures when knowledge has not been retained.<br>C20:TR-26; R21:NEW-19 | Track observed practice and unresolved learning needs using a short checklist.<br>Shared/W04 · H · Stage 1<br>n8n: Optional non-critical training reminder | **Review:** Trainer checks understanding and readiness.<br>**Evidence:** Reading or watching alone does not mark competence complete. |
| **PP-076 Owner led training burden.** New staff require substantial direct instruction and supervision.<br>R14:M1-30 | Reuse approved training materials and distribute named trainer ownership.<br>Shared/W04 · H · Stage 1<br>n8n: None required | **Review:** Management funds and provides supervision.<br>**Evidence:** Time released is measured, not assumed from a checklist. |
| **PP-077 Telephone coaching.** Staff need observed practice, recording reviews, feedback and readiness checks.<br>C20:TR-22; C20:TR-26 | Arrange authorized recording references, supervised fictional practice and objective coaching checks.<br>Shared/W04 · H · Stage 1<br>n8n: None required | **Review:** Trainer assesses telephone readiness.<br>**Evidence:** No automatic assessment authorizes independent calls. |
| **PP-078 Incomplete task sequencing.** Staff can move to another task before finishing the required updates and follow-ups.<br>C20:TR-23 | Show required steps, next owner and incomplete actions in a persistent case before closure.<br>Shared · A · Stage 1<br>n8n: None required | **Review:** Case owner verifies the actual outcome.<br>**Evidence:** An unfinished native update cannot be hidden by a sent message. |

# Appendix 14 Opportunities PP-079 to PP-084

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-079 Unverified policy understanding.** Supplying documents does not establish that staff read or understood them. R14:M1-22 | Record policy version receipt and an observed understanding check where required. Shared/W04 · H · Stage 1 n8n: None required | **Review:** HR/trainer verifies understanding. **Evidence:** Acknowledgement is not treated as proven compliant behavior. |
| **PP-080 Delayed application access.** Missing invitations, permissions, credentials or instructions prevent independent work. R14:M1-23; R21:NEW-20; C20:TR-29; C20:TR-34 | Use an account/permission checklist and verify successful authorized access. Shared/W04 · H · Stage 0 n8n: Optional identity-tool adapter only if already approved | **Review:** IT or account administrator provisions access. **Evidence:** No account is considered ready from an invitation alone. |
| **PP-081 Phone configuration work.** Office-number access, licences, caller identity, routing and voicemail require setup and confirmation of the actual account configuration. C20:TR-28 | Resolve office number, licence and routing configuration in the existing telephony account. Shared · H · Stage 0 n8n: None required | **Review:** Authorized telephony administrator configures and tests it. **Evidence:** Inbound/outbound and missed-call behavior is tested on the real account. |
| **PP-082 Group administration.** Communication groups require membership, privacy boundaries and ownership decisions. C20:TR-27 | Agree group purpose, membership and private/admin scope; record authorized ownership. W13/Shared · H · Stage 0 n8n: None required | **Review:** Office/IT manages existing communication tools. **Evidence:** A group post is not proof of task acceptance or completion. |
| **PP-083 Audio faults.** Microphone and call settings interrupt participant communication and training. R14:M1-33 | Use an IT setup check and support procedure for microphones and call settings. Shared · H · Stage 0 n8n: None required | **Review:** IT fixes hardware/application configuration. **Evidence:** A successful real call test verifies the fix. |
| **PP-084 Unclear meeting identities.** Generic accounts can make it difficult to establish who is present during sensitive discussions. R14:M1-34 | Use identifiable accounts and an attendee check before sensitive discussions. Shared · H · Stage 0 n8n: None required | **Review:** Meeting owner controls attendance and disclosure. **Evidence:** The people present are known before sensitive discussion. |

# Appendix 15 Opportunities PP-085 to PP-090

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-085 Manual absence history reconstruction.** Managers search communication and roster records to establish factual absence histories; interpretation requires human review. R14:M1-11 | Link factual absence dates, affected bookings and authorized communication references. W08/W04 · R · Stage 2 n8n: None required | **Review:** Manager interprets evidence and handles personnel review. **Evidence:** No assumption of intent becomes an automated finding. |
| **PP-086 Incomplete offboarding.** Former staff can remain on the website or payroll lists, requiring coordinated status and access review. C20:TR-07; C20:TR-32; R21:NEW-23 | Coordinate website, payroll and access offboarding as separate authorized checklist steps. W04/W11/Shared · H · Stage 1 n8n: Optional low-risk website adapter after approval | **Review:** Office, payroll and IT retain their respective decisions. **Evidence:** Published profile, run exceptions and access changes are checked separately. |
| **PP-087 Dependence on the owners.** Routine questions, exceptions, callbacks and approvals repeatedly return to the owners. C05:B05; R14:M1-25 | Use named case ownership, decision rules and exception escalation to reduce routine owner interruptions. Shared · H · Stage 1 n8n: None required | **Review:** Management delegates authority and maintains coverage. **Evidence:** Cases progress within approved authority; exceptions remain owned. |
| **PP-088 Difficulty stepping away.** Owners report that operations become less reliable when they are unavailable. C05:B05 | Provide a shared handover of open work, accountable owners and operational failures. Shared · H · Stage 1 n8n: None required | **Review:** Management supplies actual cover and authority. **Evidence:** The office can explain every open case during owner absence. |
| **PP-089 Recruitment and backup pool maintenance.** Applications require screening, practical assessment, availability checks and training; a reliable backup pool still needs people. R14:M1-28; R14:M1-29; C05:B01 | Use existing recruitment/HR tools and readiness references for backup coverage; avoid building a new applicant system. W04/W08 · H · Stage 2 n8n: None required | **Review:** Management recruits, assesses and trains workers. **Evidence:** A ready worker, not a software match, provides service capacity. |
| **PP-090 Lost service revenue.** Uncovered or cancelled services reduce revenue while recurring costs remain, according to the meeting discussion. R14:BC-01; C05:B05 | Report actual uncovered/cancelled services and verified outcomes from native data. W08/W10 · H · Stage 3 n8n: None required | **Review:** Management decides staffing and commercial response. **Evidence:** No speculative revenue recovery is counted as achieved savings. |

# Appendix 16 Opportunities PP-091 to PP-096

**Treatment:** A = bounded preparation; R = reviewed action; N = native configuration; H = human/process/IT; C = clarify first. Stages are defined in §12. No n8n dependency is required.

| Existing problem | Proposed approach | Human boundary and completion |
|---|---|---|
| **PP-091 Pressure from administration and subscriptions.** Existing costs constrain decisions about adding support staff. R14:BC-05; C05:B05 | Measure assisted handling, recovery and operating costs alongside the existing process. Shared · H · Stage 1 n8n: None required | **Review:** Owners decide staffing and subscriptions. **Evidence:** Total cost includes maintenance and staff review, not only software fees. |
| **PP-092 Limited marketing capacity.** Owners lack time or specialist knowledge to maintain consistent promotion. R14:M1-31; C05:B03 | Retain approved marketing tasks in existing tools; no campaign engine is part of the initial project. Shared · H · Stage Later separate scope n8n: Optional peripheral publishing only if separately approved | **Review:** Owner or marketer defines and executes marketing work. **Evidence:** Marketing automation remains a separate scope decision. |
| **PP-093 Variable demand.** Holidays and uneven enquiry volumes complicate staffing and growth planning. R14:BC-03 | Use existing booking/enquiry history for basic operational reports where already required. W08/Shared · H · Stage 3 n8n: None required | **Review:** Management interprets seasonal demand and staffing implications. **Evidence:** No new forecasting model or guaranteed demand prediction is introduced. |
| **PP-094 Supplier coordination.** Website changes, email licences, renewals and provider communication require management time. R14:M1-32 | Track supplier changes and renewals using existing administrative tasks/calendar. Shared · H · Stage 0 n8n: Optional non-critical calendar/reminder bridge | **Review:** Office/IT coordinates the provider work. **Evidence:** A checked supplier outcome closes the task; no migration module is built. |
| **PP-095 Memory based planning.** Holiday closures and future follow-ups depend on notes and someone remembering to raise them. R14:M1-27 | Create assigned reminders for agreed closure decisions and follow-ups, retaining the approved effective dates. Shared · A · Stage 1 n8n: Optional approved reminder delivery only | **Review:** Office approves closure policy and staffing exceptions. **Evidence:** Proposed holiday dates cannot change the roster as an approved policy. |
| **PP-096 Risk of adding duplicated work.** The client wants improvements around existing ShiftCare functions without creating another repetitive workflow. C05:P02; C05:P03 | Reuse native records, reminders, worker acceptance and financial export; store only coordination state and verified references. Shared · N · Stage 0 n8n: No n8n dependency required | **Review:** Owners confirm which gaps warrant custom functionality. **Evidence:** One authoritative owner and one write path exist for each business operation. |